Venkata Lakshmi Parimala Pasupuleti

Sr Software Engineer · Full-Stack Development · Python · React · Node.js · PostgreSQL · AWS · Docker · Kubernetes · Real Estate Domain · AI-First

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PROFESSIONAL SUMMARY

Senior Full-Stack Software Engineer with 8+ years designing, developing, testing, and deploying production applications in real estate software at scale. At Entrata (PropTech / Real Estate SaaS), owned the full SDLC for 20+ full-stack modules — Python, PHP, Java, React, Angular, PostgreSQL, RabbitMQ, AWS — serving 100K+ monthly active users at 99.5% uptime across 120+ Agile sprints. Owned a mission-critical compliance pipeline solo for 7 years without a PM or spec: zero compliance violations, zero security incidents. Mentored 5+ engineers; 300+ peer code reviews. AI-first: built AgentProbe (open-source eval toolkit, CI regression harness) and Leasel (8-agent AI platform, 93.6% precision across 1,240 test cases). M.S. Data Analytics Engineering, George Mason University (GPA 3.87). — github.com/venkatalakshmiparimala/AgentProbe

WORK EXPERIENCE

Blue Horse Digital
Sep 2026 – Present

AI/ML Engineer (Volunteer) · Remote

- Building ML models, data pipelines, and production APIs; integrating AI components across full-stack applications; debugging cross-stack reliability issues and improving software quality.

George Mason University
Aug 2024 – Aug 2025

Research Associate + Teaching Assistant

- Built multi-source ML prediction pipelines (Python, R) across 190+ countries (R^2=0.977); HuggingFace fine-tuning + inference API deployment; statistical modeling and validation. Delivered Python and SQL workshops to 600+ students across multiple sections.

Entrata (PropTech / Real Estate SaaS)
Sep 2018 – Aug 2024

Senior Software Engineer · Remote

- **Full-stack development and ownership across 20+ production modules** (Python, PHP, Java, React, Angular, PostgreSQL, RabbitMQ, AWS) serving 100K+ monthly active users at 99.5% uptime across 120+ Agile sprints — sprint planning, stand-ups, retrospectives, and continuous improvement cycles. Translated business requirements into technical solutions; communicated effectively with technical and non-technical stakeholders. Drove features from concept to production with minimal supervision. Zero compliance violations. Zero security incidents.
- **Architecture, design, and code quality** — contributed to microservices redesign using Docker and Kubernetes (50% scalability improvement, 40% deployment time reduction); 200+ SQL optimizations (30% performance gain via execution plan analysis, indexing, partitioning); RESTful API and database design across multi-tenant distributed systems; integrated applications into CI/CD pipelines (GitHub Actions) with automated testing gates. Conducted 300+ code reviews enforcing best practices and coding standards; unit tests, integration tests, and automated test coverage across all modules.
- **Document management at scale + observability** — led AWS S3 document migration for 100K+ units (100% PHI compliance, Libsodium encryption, zero security incidents); 56 state-specific document templates; instrumented systems with OpenTelemetry + Prometheus + Grafana (40% debugging time reduction); Python/Bash automation eliminating 80% of manual operational workload. Mentored 5+ junior engineers with knowledge-sharing and best practices.

Entrata India, Pune
Jul 2017 – Sep 2018

Associate Software Engineer

- Built real estate backend modules (PHP, Python, Java, PostgreSQL); led AWS S3 document migration; data pipelines across hundreds of thousands of property records; document template development with merge fields.

PROJECTS

AgentProbe: Open-Source AI Evaluation Toolkit
CI-INTEGRATED · OPEN SOURCE

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · Node.js | github.com/venkatalakshmiparimala/AgentProbe

- 15-endpoint FastAPI backend; 5-metric eval engine; LLM-as-judge scoring; CI regression harness with golden datasets; distributed model routing across OpenAI/Anthropic/Groq/Ollama. Built from scratch, prototype to production, because the tooling didn't exist.

Leasel: Full-Stack AI Investigation Platform
REAL ESTATE AI · LANGGRAPH

Python · LangGraph · FastAPI · React-compatible REST APIs · SQLite · OpenAI API | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Full-stack AI platform (FastAPI backend + RESTful APIs) with 8-agent LangGraph orchestration, durable state, RAG over document corpus, and fault-tolerant HITL escalation. 93.6% precision and 87.9% F1 across 1,240 residential lease test cases. Built iteratively with user feedback shaping the product roadmap.

EDUCATION

M.S. Data Analytics Engineering George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering JNTU Anantapur — 2017

SKILLS

Languages & Frameworks: Python (8+ yrs), JavaScript, TypeScript, Node.js, Java (Spring Boot), PHP, SQL, PL/SQL, R, Bash — React.js, Angular, FastAPI, Django, .NET Framework, GraphQL, REST APIs

Backend, Data & Infrastructure: PostgreSQL, Redis, RabbitMQ, Apache Kafka, Microservices Architecture, Distributed Systems, Event-Driven Architecture, Multi-Tenant Architecture, Idempotency, Docker, Kubernetes (K8s), Terraform, AWS (S3, EC2, RDS, Lambda, Step Functions), Azure, Google Cloud Platform, GitHub Actions CI/CD, Unit Testing, Integration Testing, Automated Testing, TDD, Code Quality, Performance Engineering

AI & Observability: LangGraph, LangChain, RAG Pipelines, LLM Evaluation, AgentProbe, Agentic Workflows, Document Management, OpenTelemetry, Prometheus, Grafana, MLflow, PyTorch, HuggingFace, Agile/Scrum, Jira